

RINGS AND FIELDS (NEP UGCF 2022, SEM VIII)

PREREQUISITES

Rings and Fields

CONTENTS

- | | |
|--|--|
| 01 §0 — What this is, who it's for, how to use | 07 §6 — Linear algebra: just enough for Units IV–VII |
| 02 §1 — The basic language: sets, functions, relations | 08 §7 — Complex numbers and roots of unity |
| 03 §2 — Group theory in 10 minutes | 09 §8 — Quick self-check (try these before opening Unit I) |
| 04 §3 — Number theory you'll keep meeting | 10 §9 — Where to learn more if a topic is rusty |
| 05 §4 — Ring theory before the syllabus starts ★ | 11 Final advice from your study notes |
| 06 §5 — Polynomial basics | |

Companion to: the 7 unit notes in this folder.

Time needed: ~6–8 hours total (single weekend if focused).

Tone: plain-language, tutor-style. Not a textbook.

§0 — What this is, who it's for, how to use

You're about to study DU's GE-8(i) Rings and Fields paper. The official syllabus starts at Herstein §3.3 — but reading those notes requires you to already know §3.1, §3.2, plus chunks of group theory, linear algebra, set theory, and number theory.

This file teaches only those bits. Nothing more, nothing less. It's the safety floor under the syllabus.

How to use this file:

If you...	Do this
Know zero algebra (your case, per what you told me)	Read top-to-bottom, in order. Don't skip sections. Try the self-check at the end.
Have a vague memory of "groups and rings" from earlier semesters	Skim section headings. Jump to anything that feels rusty. Do the self-check.
Are confident in basics	Just do §8 self-check. If you pass, open Unit I.

Notation conventions used throughout this file and the unit notes:

Symbol	Meaning
$\mathbb{Z}$	integers ..., -2, -1, 0, 1, 2, ...
$\mathbb{Q}$	rational numbers a/b with $a, b \in \mathbb{Z}$, $b \neq 0$
$\mathbb{R}$	real numbers (also used as a generic "ring" symbol — context tells you)
$\mathbb{C}$	complex numbers $a + bi$
$\mathbb{Z}_n$ or $\mathbb{Z}/n\mathbb{Z}$	integers mod n (i.e. $\{0, 1, \dots, n-1\}$)
$\in$	"is in" / "is an element of"
$\subseteq$	"is a subset of"
$\Rightarrow$	"implies"
$\Leftrightarrow$	"if and only if" / "iff"
$\exists, \forall$	"there exists", "for all"

Star ★ marks any fact that gets used constantly in the unit notes — prioritise these.

§1 — The basic language: sets, functions, relations

1.1 Functions: injective, surjective, bijective

A function $f : A \rightarrow B$ takes inputs from set A and gives outputs in set B . Each input gets exactly one output.

Property	Plain English	Example
Injective (one-to-one)	No two different inputs share the same output.	$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 2n$ is injective: if $2a = 2b$, then $a = b$.
Surjective (onto)	Every element of B is actually hit by some input.	$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 2n$ is NOT surjective: nothing maps to 3.
Bijective	Both — injective AND surjective.	$f : \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = n + 5$ is bijective.

Why this matters in the course: "isomorphism" = a structure-preserving bijection. To prove two rings are isomorphic you must show your map is both injective and surjective. (See Unit I §3.3.7, Lemma 3.3.3.)

1.2 Equivalence relations and equivalence classes ★

An equivalence relation $\sim$ on a set S is a way of saying "these two elements are the same for our purposes". It must satisfy three rules:

- 1 Reflexive: $a \sim a$ for every a .
- 2 Symmetric: $a \sim b \implies b \sim a$.
- 3 Transitive: $a \sim b$ and $b \sim c \implies a \sim c$.

Classic example. On $\mathbb{Z}$, say $a \sim b$ if $a - b$ is divisible by 5. Then:

- $7 \sim 12$ (since $12 - 7 = 5$, divisible by 5) ✓
- $7 \sim 2$ (since $7 - 2 = 5$) ✓
- $7 \sim 17$ (since $17 - 7 = 10$) ✓

All integers that are "congruent to 7 mod 5" are in one bucket. Each bucket is called an equivalence class. Here we get 5 buckets: those congruent to 0, 1, 2, 3, 4 mod 5.

Why this matters: Unit I and Unit VI both use equivalence classes in quotient-style constructions (cosets in R/U , formal fractions in the field of quotients, residues in finite-field arithmetic). For now, the take-away is the bucket idea: one class can contain many different representatives, and a function defined on classes must respect that (next section).

1.3 Well-defined functions ★

Sometimes you want to define a function whose input is an equivalence class, not a single element. You define it by picking some representative of the class and computing — but then you must check: do you get the same answer no matter which representative you picked?

Quick example. Define $f : \mathbb{Z}_6 \rightarrow \mathbb{Z}_3$ by "take any representative $n \in \mathbb{Z}$ of the class, and compute $n \bmod 3$ ".

- Is this well-defined? Take the class of 2 in $\mathbb{Z}_6$. Representatives include 2, 8, 14, $2 \bmod 3 = 2$, $8 \bmod 3 = 2$, $14 \bmod 3 = 2$. Same answer. ✓
- More generally: if $a - b$ is divisible by 6, is $a - b$ divisible by 3? Yes (6 is a multiple of 3). So same $\bmod 3$ value. Well-defined.

Whenever a later construction defines something by choosing a representative from an equivalence class, you must check that the answer does not depend on which representative you picked. That's what well-defined means. It's the most common source of "where does that step come from?" confusion.

Why this matters: in Unit I §3.4 and §3.6 you'll meet operations that are defined this way, and the unit notes spend a couple of paragraphs proving they're well-defined. When you reach those, recognise the pattern.

§2 – Group theory in 10 minutes

You don't need to be a group theory ninja for this course, but you do need the vocabulary, because the unit notes constantly say things like "ideals are to rings as normal subgroups are to groups".

2.1 What's a group?

A group is a set G with one binary operation (usually written $\cdot$ or $+$) such that:

- 1 Closure: $a \cdot b \in G$ for all $a, b \in G$.
- 2 Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- 3 Identity element: there exists $e \in G$ with $e \cdot a = a \cdot e = a$ for all a .
- 4 Inverses: for each $a \in G$, there's some $a^{-1} \in G$ with $a \cdot a^{-1} = a^{-1} \cdot a = e$.

An abelian group also has $a \cdot b = b \cdot a$ (commutativity).

Examples to keep in mind:

Group	Operation	Identity	Inverse of a	Abelian?
$(\mathbb{Z}, +)$	addition	0	$-a$	Yes
$(\mathbb{Q}^*, \cdot)$ (nonzero rationals)	multiplication	1	$1/a$	Yes
$\mathbb{Z}_n$	addition mod n	0	$n - a$	Yes
S_3 (permutations of 3 things)	composition	identity perm	reverse	No

Why this matters: every ring R is automatically an abelian group under its addition $(R, +)$. So group facts about addition apply to rings for free.

2.2 Subgroup and normal subgroup

A subgroup H of G is a subset that's itself a group under the same operation: contains the identity, closed under the operation, closed under inverses.

Example. $2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$ is a subgroup of $(\mathbb{Z}, +)$.

A normal subgroup N of G is a subgroup with the extra property $g \cdot N \cdot g^{-1} = N$ for every $g \in G$. (For abelian groups every subgroup is automatically normal — so when we say "normal subgroup of $(\mathbb{R}, +)$ ", we just mean subgroup.)

2.3 Quotient group G/N ★

If N is a normal subgroup of G , you can form a new group whose elements are the cosets $gN = \{gn : n \in N\}$. The operation is $(g_1 N)(g_2 N) = (g_1 g_2) N$.

The prototype: $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$. Take $G = \mathbb{Z}$, $N = n\mathbb{Z}$. The cosets are $n\mathbb{Z}$, $1 + n\mathbb{Z}$, $2 + n\mathbb{Z}$, ..., $(n-1) + n\mathbb{Z}$. We rename them $0, 1, 2, \dots, n-1$ and addition is "add then mod by n ". That's the group $\mathbb{Z}_n$.

Why this matters: the quotient ring R/U (Unit I §3.4) uses the same construction, with addition coming for free from the additive group structure and the new bit being multiplication.

2.4 Group homomorphism + kernel

A group homomorphism $\phi : G \rightarrow G'$ is a map satisfying $\phi(a \cdot b) = \phi(a) \cdot \phi(b)$.

The kernel is $\ker \phi = \{a \in G : \phi(a) = e'\}$ (everything mapping to the identity of G').

Key facts (use without proof in this course):

- $\ker \phi$ is always a normal subgroup of G .
- ϕ is injective $\iff \ker \phi = \{e\}$ (just the identity).

2.5 First Isomorphism Theorem for groups (statement only) ★

If $\phi : G \rightarrow G'$ is an onto group homomorphism with kernel N , then $G/N \cong G'$ (isomorphic). Don't worry about the proof — we won't reproduce it. But the ring version is Theorem 3.4.1 of the syllabus, and it's structurally identical.

2.6 Cyclic group, Lagrange, FTFAQ (one-liners)

- Cyclic group: every element is a power of a single generator g . Example: $\mathbb{Z}_n$ is cyclic (generated by 1). The multiplicative group of any finite field will turn out to be cyclic.
- Lagrange's theorem: in a finite group G , the order of any subgroup divides $|G|$. Consequence: $a^{|G|} = e$ for every a in a finite group.
- Fundamental Theorem of Finite Abelian Groups (FTFAQ): every finite abelian group is a direct sum of cyclic groups. (We use this in Unit VI §B.1 to show $(GF(p^n), +) \cong \mathbb{Z}_{p^n}$.)

You will not be asked to prove these. You just need to recognise the names when the unit notes invoke them.

§3 – Number theory you'll keep meeting

3.1 Divisibility

$a \mid b$ means "a divides b", i.e. $b = ac$ for some integer c . Read it: "a divides b".

- $3 \mid 12$ ✓ ($12 = 3 \cdot 4$)
- $5 \mid 12$ ✗
- $a \mid 0$ for every a ($0 = a \cdot 0$).

3.2 Greatest common divisor (gcd) ★

$\gcd(a, b)$ is the largest positive integer dividing both a and b . Standard ways:

- By inspection: $\gcd(12, 18) = 6$.
- By prime factorisation: $12 = 2^2 \cdot 3$, $18 = 2 \cdot 3^2$. Take min of each prime exponent: $\gcd = 2 \cdot 3 = 6$.
- By Euclidean algorithm (next).

3.3 Euclidean algorithm (worked) ★

To compute $\gcd(48, 18)$:

```

48 = 2 · 18 + 12    ← remainder 12
18 = 1 · 12 + 6     ← remainder 6
12 = 2 · 6 + 0      ← remainder 0, stop
gcd(48, 18) = 6     ← last nonzero remainder

```

The extended Euclidean algorithm also gives you integers λ, μ with $\gcd(a, b) = \lambda a + \mu b$ (Bézout). Tracing the above backwards:

```

6 = 18 - 1 · 12
  = 18 - 1 · (48 - 2 · 18)
  = 3 · 18 - 1 · 48

```

So $\gcd(48, 18) = 6 = (-1) \cdot 48 + 3 \cdot 18$. Verify: $-48 + 54 = 6$. ✓

Why this matters: Unit II §3.7 generalises both the algorithm and Bézout from $\mathbb{Z}$ to any "Euclidean ring". The integer version is the prototype to keep in mind.

3.4 Primes and prime factorisation

A prime in $\mathbb{Z}$ is a positive integer $p > 1$ whose only divisors are ± 1 and $\pm p$. The first few: 2, 3, 5, 7, 11, 13, 17, 19, 23, ...

Fundamental Theorem of Arithmetic. Every integer > 1 factors uniquely into primes (up to order). Example: $360 = 2^3 \cdot 3^2 \cdot 5$.

Why this matters: Unit II's "Unique Factorisation Theorem" (Thm 3.7.2) generalises this from $\mathbb{Z}$ to any Euclidean ring.

3.5 Modular arithmetic and $\mathbb{Z}_n$

$a \bmod n$ is the remainder when a is divided by n . $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ with addition and multiplication "mod n ":

- In $\mathbb{Z}_6$: $4 + 5 = 9 \equiv 3$, so $4 + 5 = 3$ in $\mathbb{Z}_6$. Also $3 \cdot 4 = 12 \equiv 0$, so $3 \cdot 4 = 0$ in $\mathbb{Z}_6$ (a "zero divisor" — see §4).
- In $\mathbb{Z}_5$: $3 \cdot 2 = 6 \equiv 1$, so $2^{-1} = 3$ in $\mathbb{Z}_5$. Every nonzero element has an inverse iff n is prime — so $\mathbb{Z}_p$ is a field.

3.6 Fermat's Little Theorem

If p is a prime and a is not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}, \text{ equivalently } a^p \equiv a \pmod{p}.$$

The second form ($a^p = a$ in $\mathbb{Z}_p$) holds for all a including $a = 0$.

Why this matters: Unit VI proof of Theorem 22.1 uses $\alpha^{p^n} = \alpha$ for every $\alpha \in GF(p^n)$, which is a "big sister" of Fermat. Also: every element $c \in \mathbb{Z}_p$ satisfies $c^p = c$ — this is used in Unit VI §B.5 (the "Why Frobenius sends roots to roots" paragraph).

3.7 The "small binomial coefficient is divisible by p " trick ★

For any prime p and integer k with $0 < k < p$:

$$\binom{p}{k} = \frac{p!}{k!(p-k)!} \text{ is divisible by } p.$$

Quick proof. The numerator $p!$ has the factor p . The denominator $k!(p-k)!$ has no factor of p (because both $k < p$ and $p - k < p$, so neither factorial contains p). So p survives in the integer quotient $\binom{p}{k}$.

Consequence — the Frobenius identity: in any commutative ring of characteristic p (or whenever a and b commute in a characteristic- p ring),

$$(a + b)^p = a^p + b^p.$$

(Because all middle terms $\binom{p}{k} a^k b^{p-k}$ are multiples of p , hence zero in characteristic p . Commutativity is needed for the binomial theorem to hold — otherwise the expansion picks up cross terms.)

Why this matters: this is THE engine behind every "Frobenius" argument in Units IV (Thm 20.7) and VI (Thm 22.1, Thm 22.3). Memorise the trick.

§4 – Ring theory before the syllabus starts ★

This is the biggest section. Herstein §3.1–§3.2. The actual syllabus starts at §3.3, but every page of every unit assumes you know this.

4.1 What's a ring?

A ring R is a set with two operations $+$ and $\cdot$ satisfying:

Addition axioms: $(R, +)$ is an abelian group. So:

- closure under $+$
- associative
- has 0 (additive identity)
- has $-a$ for each a (additive inverse)
- commutative: $a + b = b + a$

Multiplication axioms:

- closure under $\cdot$
- associative

Mixing axioms (distributive laws):

- $a \cdot (b + c) = a \cdot b + a \cdot c$
- $(b + c) \cdot a = b \cdot a + c \cdot a$

That's the definition. Note multiplication does NOT need to be commutative, does NOT need an identity, and elements do NOT need to have multiplicative inverses.

4.2 The big examples (memorise these)

Ring	Operations	Comm?	Has 1?	Integral domain?	Field?
$\mathbb{Z}$	usual $+, \cdot$	✓	✓ (yes, 1)	✓	✗ (2 has no inverse)
$\mathbb{Q}, \mathbb{R}, \mathbb{C}$	usual $+, \cdot$	✓	✓	✓	✓
$\mathbb{Z}_n$	$+, \cdot \bmod n$	✓	✓	iff n prime	iff n prime
$\mathbb{Z}[i]$ (Gaussian integers)	usual $+, \cdot$ of $a + bi$	✓	✓ (yes, 1)	✓	✗
$M_n(\mathbb{R})$ ($n \times n$ real matrices)	matrix $+, \cdot$	✗ (for $n \geq 2$)	✓ (identity matrix)	✗	✗
$F[x]$ (polys over field F)	usual poly $+, \cdot$	✓	✓	✓	✗

Ring	Operations	Comm?	Has 1?	Integral domain?	Field?
$C[0,1]$ (cts real-valued fns on $[0,1]$)	pointwise $+$, $\cdot$	✓	✓ (constant function 1)	✗ ($f \cdot g = 0$ possible without $f = 0$)	✗

The terms in the column headers are defined below.

4.3 Commutative ring; ring with unity ★

- Commutative ring: $a \cdot b = b \cdot a$ for all a, b .
- Ring with unity (or "ring with 1"): there's an element $1 \in R$ with $1 \cdot a = a \cdot 1 = a$ for every a .

Most rings you meet in this course are both. $M_n(R)$ is the famous exception (non-commutative).

4.4 Subring

A subring is a subset $S \subseteq R$ that is itself a ring under the same operations: closed under $+$, $\cdot$, $-$ (additive inverses). Some books also demand S contain 1 ; Herstein does not.

- $\mathbb{Z}$ is a subring of $\mathbb{Q}$, of $\mathbb{R}$, of $\mathbb{C}$.
- $\mathbb{Z}[i]$ is a subring of $\mathbb{C}$.
- The set of even integers $2\mathbb{Z}$ is a subring of $\mathbb{Z}$.

Non-example. $\{0, 1\} \subseteq \mathbb{Z}$ is not a subring: it's not closed under $+$ (since $1 + 1 = 2 \notin \{0, 1\}$), and it's not closed under additive inverses either (since $-1 \notin \{0, 1\}$). Mere "contains 0 and 1" is not enough.

(Don't confuse subring with ideal — that's a much stronger notion you'll meet in Unit I §3.4.)

4.5 Zero divisors ★

In some rings, two nonzero things can multiply to zero. Such elements are called zero divisors:

$a \neq 0$ is a zero divisor if there's some $b \neq 0$ with $ab = 0$ (or $ba = 0$).

Examples:

- In $\mathbb{Z}_6$: $2 \cdot 3 = 6 \equiv 0$, so both 2 and 3 are zero divisors.
- In $\mathbb{Z}_8$: $4 \cdot 2 = 8 \equiv 0$, so 4 and 2 are zero divisors.
- In $M_2(\mathbb{R})$: zero divisors exist — pick any singular matrix.
- In $C[0,1]$: take $f(x) = \max(0, x - 1/2)$ and $g(x) = \max(0, 1/2 - x)$. Both nonzero, but $f \cdot g = 0$ everywhere.

Rings with no zero divisors are called integral domains (next).

4.6 Integral domain ★

An integral domain is a commutative ring with unity $1 \neq 0$ and no zero divisors.

Equivalent characterisation: $ab = 0 \implies a = 0$ or $b = 0$.

A useful consequence: cancellation law. In an integral domain, if $ac = bc$ and $c \neq 0$, then $a = b$ (subtract: $(a - b)c = 0$, no zero divisors, $a - b = 0$).

Examples: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}[i]$, $\mathbb{Z}_p$ for prime p , $F[x]$ for any field F . Non-examples: $\mathbb{Z}_6$, $M_2(\mathbb{R})$, $\mathbb{C}[0,1]$.

4.7 Units (invertible elements)

A unit in a ring with unity R is an element u with a multiplicative inverse: there exists $v \in R$ with $uv = vu = 1$.

- In $\mathbb{Z}$: units are ± 1 .
- In $\mathbb{Z}_n$: units are integers coprime to n . So in $\mathbb{Z}_6$: units are $\{1, 5\}$ (since $5 \cdot 5 = 25 \equiv 1$). In $\mathbb{Z}_7$ (p prime): units are $\{1, 2, 3, 4, 5, 6\}$, all nonzero.
- In $\mathbb{Z}[i]$: units are $\{1, -1, i, -i\}$ (norms equal to 1).
- In $F[x]$ (F a field): units are nonzero constants.

The set of units of R , denoted $U(R)$, is a group under multiplication (closed, has 1, has inverses). Sometimes called the group of units.

Warning. Don't confuse the "unit" (an element with an inverse) with the "unit element" 1 (the multiplicative identity). Unfortunate terminology collision.

4.8 Field ★

A field F is a commutative ring with unity $1 \neq 0$ in which every nonzero element is a unit. I.e., you can always divide by anything except 0.

Examples: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ for prime p . Non-examples: $\mathbb{Z}$, $\mathbb{Z}_6$, $F[x]$ (you can't divide polynomials in general).

Tiny fact (used silently in many proofs): every field is an integral domain. (A unit can't be a zero divisor: if $u \cdot b = 0$ and u^{-1} exists, multiply by u^{-1} to get $b = 0$.)

4.9 Characteristic of a ring ★

The characteristic $\text{char } R$ of a ring with unity is the smallest positive integer n such that $n \cdot 1 = 1 + 1 + \dots + 1$ (n copies) $= 0$. If no such n exists, $\text{char } R = 0$.

Ring	Characteristic
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	0
$\mathbb{Z}_n$	n
$\mathbb{Z}_p$ for prime p	p
$F[x]$ for F a field	$\text{char } F$

Important fact (used in Unit IV): if R is an integral domain (in particular, a field), then $\text{char } R$ is either 0 or a prime number. (If $\text{char } R = n = ab$ with $1 < a, b < n$, then $(a \cdot 1)(b \cdot 1) = ab \cdot 1 = 0$, giving zero divisors — contradiction.)

4.10 Quick comparison table

Property	Commutative ring with unity	Integral domain	Field
Commutative?	✓	✓	✓
Has 1?	✓	✓ (and $1 \neq 0$)	✓ (and $1 \neq 0$)
Zero divisors?	maybe	NO	NO
Every nonzero is a unit?	maybe	maybe (e.g. $\mathbb{Z}$ – no!)	YES

So: field $\implies$ integral domain $\implies$ commutative ring with unity. The reverse implications all fail.

§5 – Polynomial basics

5.1 What's a polynomial?

A polynomial in one variable x over a ring R is a formal sum

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

where the coefficients $a_i \in R$, and only finitely many are nonzero. The set of all such is $R[x]$ (read: "R adjoin x " or "polys over R ").

5.2 Addition and multiplication

Addition: add coefficient-wise.

$$(2 + 3x + x^2) + (1 + x) = 3 + 4x + x^2.$$

Multiplication: distribute and collect, using $x^a \cdot x^b = x^{a+b}$.

$$(1 + 2x)(1 - x) = 1 + 2x - x - 2x^2 = 1 + x - 2x^2.$$

Tip for $\mathbb{Z}_n[x]$: reduce coefficients mod n as you go. E.g., in $\mathbb{Z}_5[x]$: $(3x)(2x) = 6x^2 = 1 \cdot x^2 = x^2$.

5.3 Degree, leading coefficient, constant term

- Degree $\deg f(x)$ = the largest i such that $a_i \neq 0$.
- Leading coefficient = a_n where $n = \deg f$.
- Constant term = a_0 .

$\deg(3 - 2x + 7x^4) = 4$, leading coefficient 7, constant term 3.

Degree of zero polynomial: undefined (or sometimes $-\infty$). Be careful when invoking formulas like $\deg(fg) = \deg f + \deg g$ – they only hold for nonzero f, g .

5.4 Roots / zeros

A root (or zero) of $f(x)$ is an element α with $f(\alpha) = 0$. Equivalently (over a field), $(x - \alpha)$ is a factor of $f(x)$.

Examples:

- $f(x) = x^2 - 2$ has roots $\pm\sqrt{2} \in \mathbb{R}$.
- $f(x) = x^2 + 1$ has roots $\pm i \in \mathbb{C}$; over $\mathbb{R}$ it has no roots.
- $f(x) = x^2 + x + 1$ has no root in $\mathbb{Z}_2$ (check: $f(0) = 1, f(1) = 1$); has roots in some extension (which Unit IV constructs).

5.5 Rational root test ★

If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ has integer coefficients and $r/s \in \mathbb{Q}$ is a rational root (with r/s in lowest terms), then:

$$r \mid a_0 \text{ and } s \mid a_n.$$

Worked example. $f(x) = 2x^3 - 3x^2 - 11x + 6$. Candidates: $r \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$, $s \in \{\pm 1, \pm 2\}$, giving $r/s \in \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 1/2, \pm 3/2\}$.

Trial $x = 3$: $f(3) = 54 - 27 - 33 + 6 = 0$ ✓. So 3 is a root, and $f(x) = (x - 3) \cdot q(x)$ for some quadratic $q(x)$.

Why this matters: Unit III uses the rational root test to verify irreducibility of polynomials. Unit VII uses it for the trisection-of- 60° proof ($8x^3 - 6x - 1$ has no rational root, hence no linear factor, hence irreducible).

5.6 Polynomial long division (preview)

Same idea as long division of numbers, but with polynomials. Quick example: divide $x^3 + x + 1$ by $x + 1$:

$$\begin{array}{r}
 x^2 - x + 2 \\
 x + 1 \overline{) x^3 + 0x^2 + x + 1} \\
 \underline{x^3 + x^2} \\
 -x^2 + x \\
 \underline{-x^2 - x} \\
 2x + 1 \\
 \underline{2x + 2} \\
 -1
 \end{array}$$

So $x^3 + x + 1 = (x + 1)(x^2 - x + 2) + (-1)$. Quotient $x^2 - x + 2$, remainder -1 .

Why this matters: Unit III §3.9 makes this the foundation of $F[x]$ being a "Euclidean ring".

§6 — Linear algebra: just enough for Units IV–VII

6.1 Vector space over a field ★

A vector space over a field F is a set V of "vectors" with:

- vector addition $+$ (making V an abelian group),
- scalar multiplication $\cdot : F \times V \rightarrow V$ (multiplying a scalar from F by a vector in V),

satisfying the usual distributive/associative axioms (you've seen them in school: $c(v + w) = cv + cw$, $(c_1 + c_2)v = c_1v + c_2v$, $1 \cdot v = v$, etc.).

Examples for this course:

Vector space	Field	"Vectors" are...
$\mathbb{R}^2$	$\mathbb{R}$	pairs (x, y)
$\mathbb{C}$ viewed as $\mathbb{R}$ -vector space	$\mathbb{R}$	complex numbers $a + bi$, scalars are reals
$Q(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$	$\mathbb{Q}$	elements of $Q(\sqrt{2})$, scalars are rationals
Any field extension $E \supseteq F$	F	elements of E , scalars are from F

The last example is the key one for Units IV–VII: field extensions are automatically vector spaces over the smaller field.

6.2 Linear combination, linear independence

A linear combination of vectors $v_1, \dots, v_k$ (with scalars from F) is anything of the form $c_1 v_1 + c_2 v_2 + \dots + c_k v_k$.

Vectors $v_1, \dots, v_k$ are linearly independent if the only way to get the zero vector as a linear combination is the trivial one (all $c_i = 0$).

Example: in $\mathbb{R}^2$, the vectors $(1, 0)$ and $(0, 1)$ are linearly independent. Adding $(1, 1)$ to the list makes it linearly dependent: $(1, 1) - (1, 0) - (0, 1) = 0$.

6.3 Span and basis ★

The span of $v_1, \dots, v_k$ is the set of all their linear combinations: $\text{Span}(v_1, \dots, v_k) = \{c_1 v_1 + \dots + c_k v_k : c_i \in F\}$.

A basis of V is a list of vectors that is:

- 1 linearly independent, AND
- 2 spans all of V .

So every element of V is uniquely expressible as a linear combination of basis vectors.

Example. $\{(1, 0), (0, 1)\}$ is a basis of $\mathbb{R}^2$ over $\mathbb{R}$. $\{1, i\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$.

6.4 Dimension ★

The dimension of V over F , written $\dim_F V$, is the number of vectors in any basis. (Fact, stated without proof: all bases have the same number of vectors.)

- $\dim_{\mathbb{R}} \mathbb{R}^2 = 2$.
- $\dim_{\mathbb{R}} \mathbb{C} = 2$ (basis $\{1, i\}$).
- $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$ (basis $\{1, \sqrt{2}\}$).

Why this matters: in the course, $[E : F]$ (called the "degree of the extension") is defined as $\dim_F E$. The whole tower law $[K : F] = [K : E] \cdot [E : F]$ is a counting argument with bases. (See Unit V Thm 21.5.)

§7 – Complex numbers and roots of unity

7.1 Complex numbers

$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}$. Operations:

- $(a + bi) + (c + di) = (a + c) + (b + d)i$
- $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$

Conjugation: $\overline{a + bi} = a - bi$. Key property: $(a + bi)(a - bi) = a^2 + b^2$ (a non-negative real number, called the norm-squared).

Why this matters: Unit II §3.8 uses conjugation to prove $\mathbb{Z}[i]$ is a Euclidean ring (the famous "conjugate trick"). Unit IV examples use $\mathbb{C}$ as the playground.

7.2 Roots of unity ★

An n -th root of unity is a complex number ω with $\omega^n = 1$. There are exactly n of them: $\omega_k = e^{2\pi i k/n} = \cos(2\pi k/n) + i \sin(2\pi k/n)$ for $k = 0, 1, \dots, n-1$.

For $n = 3$: cube roots of unity are $1, \omega, \omega^2$ where $\omega = e^{2\pi i/3} = (-1 + i\sqrt{3})/2$. They satisfy $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.

The n -th roots of unity form a cyclic group of order n under multiplication.

Why this matters: Unit IV Example 8 — the splitting field of $x^n - a$ over $\mathbb{Q}$ involves roots of unity.

7.3 The Frobenius identity in characteristic p ★

We proved this in §3.7 above. Just restating it for emphasis: in any commutative ring of characteristic p (a prime),

$$(a + b)^p = a^p + b^p.$$

Iterating: $(a + b)^{p^n} = a^{p^n} + b^{p^n}$ (apply Frobenius n times). All the rings you'll meet in this course (Units IV–VI use it heavily on fields) are commutative, so the hypothesis is automatic.

This single identity drives Theorem 20.7 (finite fields are perfect), Theorem 22.1 (classification of finite fields), Theorem 22.3 (subfields), and the Frobenius-orbit argument throughout Unit VI. Without this fact, half of Unit IV–VI is unmotivated.

§8 – Quick self-check (try these before opening Unit I)

If you can answer all 10 in under 30 minutes without flipping pages, you're ready. If you struggle on 3+, re-read the relevant section above.

Solutions immediately after — don't peek until you've attempted.

Questions

- Q1. Define a ring. (No looking — say it out loud or write it from memory.)
- Q2. Give one example each of: (a) a commutative ring with unity, (b) a non-commutative ring with unity, (c) a ring with zero divisors, (d) a field.
- Q3. Is $\mathbb{Z}_8$ an integral domain? Justify in one line.
- Q4. Find $\gcd(56, 21)$ using the Euclidean algorithm. Express it as $56\lambda + 21\mu$ for some integers λ, μ .
- Q5. What is the characteristic of $\mathbb{Z}_5$? Of $\mathbb{Z}$? Of $\mathbb{Q}$?
- Q6. True or false: every field is an integral domain. Justify in one line.
- Q7. In $\mathbb{Z}_7$, find the multiplicative inverse of 4.
- Q8. What is $\dim_{\mathbb{R}} \mathbb{C}$? Give a basis.
- Q9. Use the rational root test to find all rational roots of $f(x) = x^3 - 4x^2 + x + 6$.
- Q10. In a commutative ring of characteristic 3, what is $(a + b)^3$ equal to? Why?

Solutions

- S1. A set with two operations $+$, $\cdot$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $+$ distributes over $\cdot$ on both sides. (See §4.1.)
- S2. (a) $\mathbb{Z}$. (b) $M_2(\mathbb{R})$. (c) $\mathbb{Z}_6$ (e.g. $2 \cdot 3 = 0$). (d) $\mathbb{Q}$. (Many other answers are correct.)
- S3. No. $2 \cdot 4 = 8 \equiv 0 \pmod{8}$ and both $2, 4 \neq 0$. So $\mathbb{Z}_8$ has zero divisors, hence is not an integral domain.
- S4. Euclidean:

$$\begin{aligned} 56 &= 2 \cdot 21 + 14 \\ 21 &= 1 \cdot 14 + 7 \\ 14 &= 2 \cdot 7 + 0 \\ \gcd &= 7 \end{aligned}$$

Back-substitute: $7 = 21 - 14 = 21 - (56 - 2 \cdot 21) = 3 \cdot 21 - 1 \cdot 56$. So $\lambda = -1, \mu = 3$.

S5. $\text{char } \mathbb{Z}_5 = 5$. $\text{char } \mathbb{Z} = 0$. $\text{char } \mathbb{Q} = 0$.

S6. True. In a field, every nonzero element has an inverse, hence cannot be a zero divisor (multiply by inverse to get $0 = \text{inverse of nonzero} \times 0 = \text{anything}$, forcing the "other" factor to be

zero).

S7. 4^{-1} in $\mathbb{Z}_7$: try $4 \cdot 1 = 4$, $4 \cdot 2 = 8 \equiv 1$. So $4^{-1} = 2$.

S8. $\dim_{\mathbb{R}} \mathbb{C} = 2$. Basis: $\{1, i\}$.

S9. Candidates: $r \mid 6$, $s \mid 1$, so $r/s \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$. Check:

- $f(1) = 1 - 4 + 1 + 6 = 4 \neq 0$.
- $f(-1) = -1 - 4 - 1 + 6 = 0 \checkmark$
- $f(2) = 8 - 16 + 2 + 6 = 0 \checkmark$
- $f(3) = 27 - 36 + 3 + 6 = 0 \checkmark$
- (± 6 won't give 0 since the polynomial has only 3 roots in $\mathbb{C}$ and we've already found 3.)

Rational roots: $-1, 2, 3$.

S10. $(a + b)^3 = a^3 + b^3$. Because $\binom{3}{1} = \binom{3}{2} = 3 \equiv 0 \pmod{3}$, so the middle binomial terms vanish in characteristic 3.

If you got 8/10 or more right: open Unit I, you're ready.

§9 — Where to learn more if a topic is rusty

The prereq notes above are deliberately compressed. If you need more depth on a topic, here are pointers (free, Indian-undergraduate-friendly):

YouTube playlists (Indian English, exam-oriented)

- Group theory basics: "Group Theory by Dr. Gajendra Purohit" or "Abstract Algebra by Mathematicia" on YouTube. Watch the first 5–8 videos for groups, subgroups, normal subgroups, quotient groups.
- Ring theory basics (§3.1–§3.2 of Herstein): "Dr. Gajendra Purohit Ring Theory" playlist (videos 1–8 are the basics — these are exactly the prereq videos for Unit I). You already have this playlist (see Q&A earlier).
- Linear algebra basics: "3Blue1Brown — Essence of Linear Algebra" (visual, 3 hours total, English with subtitles).
- Number theory + Fermat's Little Theorem: "Khan Academy: Modular arithmetic" series.

Textbooks (chapters to read)

- **Herstein, Topics in Algebra, 2nd ed.**: §3.1–§3.2 (pp. 119–131) for ring basics. Chapter 2 §2.1–§2.7 for group theory basics.
- **Gallian, Contemporary Abstract Algebra, 9th ed.**: Chapter 0 (sets, induction, divisibility, gcd, modular arithmetic), Chapter 1 (basics), Chapters 6–10 (groups), Chapter 19 (vector spaces).

When to ask vs when to read

- If you can't recall a definition → look it up in this prereq file or Gallian Ch 0–4.
- If you understand the definition but can't follow a proof in the unit notes → read the textbook proof in Herstein/Gallian at the cited page (every theorem in the unit notes cites its source page).
- If both fail → ask for clarification — but be specific about which line in which proof is the sticking point.

Final advice from your study notes

You don't need to be perfect on the prereqs above to start Unit I. You need to be comfortable enough with the vocabulary and basic constructions that when the unit notes say "let R be a commutative ring with unity and U an ideal", you don't pause to remember what a ring is.

Aim for "I can read the sentence without my brain bouncing off any of the words." That's the bar.

Once you hit that bar — open Unit I and trust the notes. They take you the rest of the way.

Good luck. ☐